

```
import pandas as pd

import numpy as np

import seaborn as sns

import matplotlib.pyplot as plt


# -----

# Load Dataset

# -----

df = pd.read_csv('/content/titanic.csv') # Original incomplete dataset
df = sns.load_dataset('titanic') # Load complete Titanic dataset


# -----

# Basic Checks

# -----

print(df.head())
print(df.tail())
print(df.info())


# Clean column names (safe step)
df.columns = df.columns.str.strip()


# Handle missing 'age' values by dropping rows
df.dropna(subset=['age'], inplace=True)


# -----

# 1. DATA VISUALIZATION (PATTERN FINDING)

# -----


# Age distribution

plt.figure()

sns.histplot(df['age'], bins=30, kde=True)
```

```
plt.title('Age Distribution')
```

```
plt.show()
```

```
# Age vs Fare relationship
```

```
plt.figure()
```

```
sns.jointplot(x='age', y='fare', data=df, kind='scatter')
```

```
plt.show()
```

```
# Count of passengers by gender
```

```
plt.figure()
```

```
sns.countplot(x='sex', data=df)
```

```
plt.title('Count of Male and Female Passengers')
```

```
plt.show()
```

```
# Age distribution by gender
```

```
plt.figure()
```

```
sns.boxplot(x='sex', y='age', data=df)
```

```
plt.title('Age Distribution by Gender')
```

```
plt.show()
```

```
# Age vs Gender vs Survival
```

```
plt.figure()
```

```
sns.boxplot(x='sex', y='age', data=df, hue='survived')
```

```
plt.title('Age vs Gender vs Survival')
```

```
plt.show()
```

```
# Violin plot
```

```
plt.figure()
```

```
sns.violinplot(x='sex', y='age', data=df, hue='survived')
```

```
plt.title('Violin Plot (Age, Gender, Survival)')
```

```
plt.show()
```

```
# Strip plot
plt.figure()
sns.stripplot(x='sex', y='age', data=df, hue='survived', jitter=True)
plt.title('Strip Plot (Age Distribution with Survival)')
plt.show()
```

```
# Swarm plot
plt.figure()
sns.swarmplot(x='sex', y='age', data=df, hue='survived')
plt.title('Swarm Plot (Survival Analysis)')
plt.show()
```

```
# Correlation heatmap
plt.figure()
corr = df.corr(numeric_only=True)
sns.heatmap(corr, annot=True)
plt.title('Correlation Heatmap')
plt.show()
```

```
# -----
# 2. REQUIRED: Fare Distribution
# -----
plt.figure()
sns.histplot(df['fare'], bins=40, kde=True)
plt.title('Fare Distribution Histogram')
plt.show()
```